1 Grammar

R R^2 $R_{\mu\nu}R^{\mu\nu}$

$\phi_1 R$ $(\nabla\phi_2)^2$ R^3

$\nabla_\mu R \nabla^\mu R$ $\cdots$

141 operators, mass dim ≤ 6

2 LLM proposal

$\mathscr{L}_1$
$\mathscr{L}_2$
$\mathscr{L}_3$
history

#2

returns index + rationale

3 Build & sample

$$\mathcal{L} = M_p^2(c_1 R + c_2 \Lambda^{-2} R^2)$$
$$-\tfrac{1}{2}(\nabla\phi_1)^2 - \tfrac{1}{2}m_1^2\phi_1^2$$

$\theta = (H_0,\ \Omega_{m0},\ \lambda_H,\ \varphi_{1,0},\ \mu_1,\ c_I)$

1000 samples $\theta \sim q_t(\theta \mid \mathcal{L})$

4 FLRW reduction

$$ds^2 = -N^2 dt^2 + a^2 d\vec{x}^2$$

$$E_N = 0, \quad E_a = 0, \quad E_{\phi_i} = 0$$

$$\mathcal{A}_{AB}^{(r)} = 0 \quad (r \geq 3)$$

remove 3rd+ time derivatives

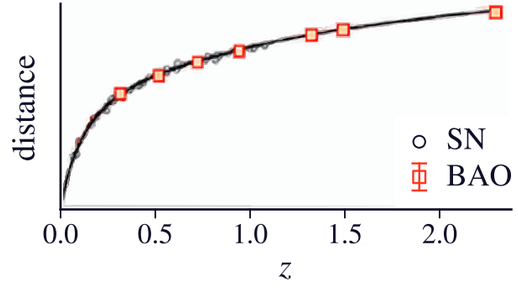
5 Solver

$$\ddot{a} = F_a(a, \dot{a}, \phi_i, \dot{\phi}_i)$$
$$\ddot{\phi}_i = F_{\phi_i}(a, \dot{a}, \phi_i, \dot{\phi}_i)$$
$$C_N(a, \dot{a}, \phi_i, \dot{\phi}_i) = 0$$

(constraint fixes $\dot{\phi}_{i,0}$; residual monitored)

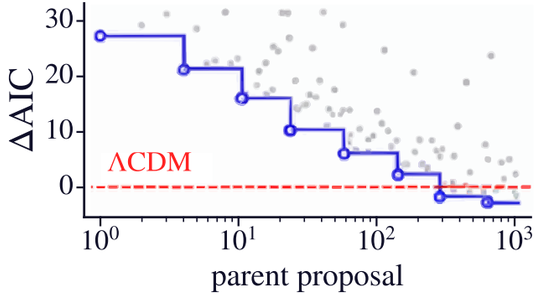
integrate ODEs $\Rightarrow H(z)$

6 Data fit



$\chi^2_{\rm SN} + \chi^2_{\rm BAO}$

7 Run memory



gray: discarded, purple: kept

next proposal

AI-driven

Deterministic physics

Data likelihood